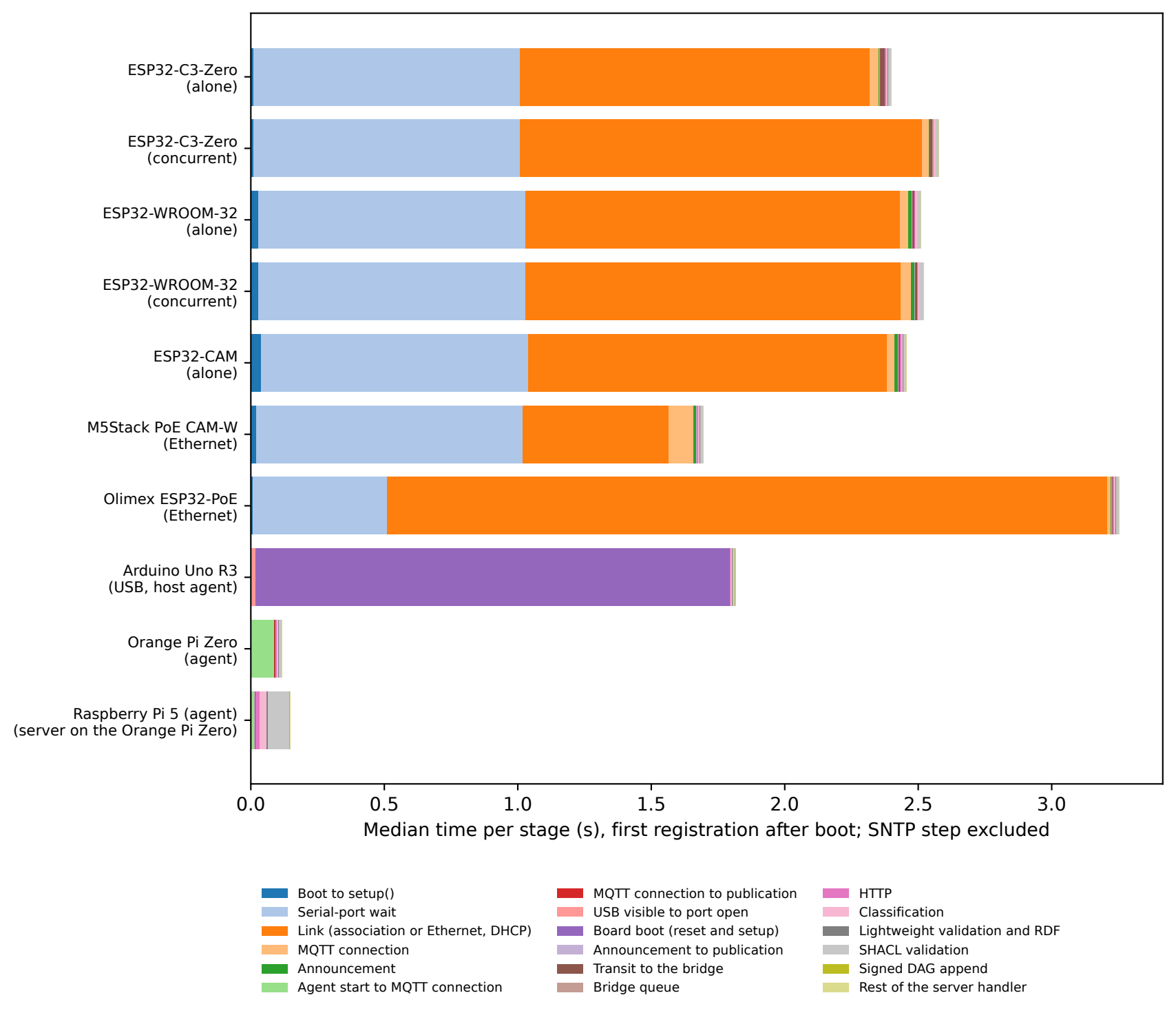
ESP32-C3-Zero (alone)
ESP32-C3-Zero (concurrent)
ESP32-WROOM-32 (alone)
ESP32-WROOM-32 (concurrent)
ESP32-CAM (alone)
M5Stack PoE CAM-W (Ethernet)
Olimex ESP32-PoE (Ethernet)
Arduino Uno R3 (USB, host agent)
Orange Pi Zero (agent)
Raspberry Pi 5 (agent) (server on the Orange Pi Zero)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Median time per stage (s), first registration after boot; SNTP step excluded
Boot to setup()
Serial-port wait
Link (association or Ethernet, DHCP)
MQTT connection
Announcement
Agent start to MQTT connection
MQTT connection to publication
USB visible to port open
Board boot (reset and setup)
Announcement to publication
Transit to the bridge
Bridge queue
HTTP
Classification
Lightweight validation and RDF
SHACL validation
Signed DAG append
Rest of the server handler